

Interview Prep — Q&A by Subject

Based on your tracker: Programming+OOPs, CN, OS, DBMS, DSA, ML, Deep Learning, Agentic AI

1. Programming + OOPs

Q1: What are the four pillars of OOP? A: Encapsulation (bundling data + methods, hiding internal details), Abstraction (showing only essential features, hiding complexity), Inheritance (a class reusing/extending another class), Polymorphism (same function name behaving differently — e.g., method overriding/overloading).

Q2: Difference between a class and an object? A: A class is a blueprint/template. An object is a real instance created from that blueprint, with its own actual data.

Q3: What is the difference between overloading and overriding? A: Overloading = same method name, different parameters, within the same class (compile-time). Overriding = a child class redefines a method already defined in the parent class (runtime).

Q4: What is a constructor? A: A special method automatically called when an object is created, used to initialize the object's data.

Q5: Is Python purely object-oriented? A: Python supports OOP but isn't purely OOP like Java — you can also write procedural-style code (functions outside any class) freely.

Q6: What is encapsulation, with an example? A: Restricting direct access to an object's internal data, usually by making variables private and exposing controlled access through methods (getters/setters). Example: a `BankAccount` class hides the `balance` variable and only lets you change it through a `deposit()` or `withdraw()` method.

Q7: What is method overriding used for in real projects? A: When a child class needs to provide its own specific behavior for a method already defined in the parent class — e.g., a `Dog` class overriding a generic `Animal.makeSound()` method to bark instead of a generic sound.

Q8: What is the difference between `is` and `==` in Python? A: `==` checks if two values are equal. `is` checks if two variables point to the exact same object in memory.

Q9: What are `*args` and `**kwargs` in Python? A: `*args` lets a function accept any number of positional arguments as a tuple. `**kwargs` lets it accept any number of keyword arguments as a dictionary — useful for writing flexible functions.

Q10: What is exception handling, and why is it important? A: Using `try/except` blocks to catch and handle runtime errors gracefully instead of letting the program crash — important for building reliable applications, like handling a failed API call in a chatbot.

2. Computer Networks (CN)

Q1: What is the difference between TCP and UDP? A: TCP is connection-oriented, reliable, ensures ordered delivery (used for web, email). UDP is connectionless, faster, no delivery guarantee (used for video streaming, gaming).

Q2: What happens when you type a URL and press Enter? A: DNS resolves the domain to an IP → browser opens a TCP connection (handshake) → sends an HTTP request → server responds → browser renders the page.

Q3: What is an IP address vs a MAC address? A: IP address is a logical address used for routing across networks. MAC address is a physical, hardware-level address unique to a network device.

Q4: What is the OSI model, briefly? A: A 7-layer conceptual model for how data travels across a network: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q5: What is DNS? A: Domain Name System — translates human-readable domain names (like google.com) into IP addresses computers use to identify each other.

Q6: What is HTTP vs HTTPS? A: HTTP transfers data in plain text. HTTPS adds encryption (via SSL/TLS) so data between browser and server can't be easily intercepted or read.

Q7: What is a firewall? A: A security system that monitors and controls incoming/outgoing network traffic based on defined rules, acting as a barrier between trusted and untrusted networks.

Q8: What is the difference between a router and a switch? A: A switch connects devices within the same local network and forwards data based on MAC addresses. A router connects different networks together and forwards data based on IP addresses.

Q9: What is a REST API, and how does it relate to your projects? A: An architectural style for web services using standard HTTP methods (GET, POST, PUT, DELETE) to communicate. My chatbot projects use REST APIs to send requests to and receive responses from the LLM (like the Gemini API).

Q10: What is latency vs bandwidth? A: Latency is the delay before data starts transferring (response time). Bandwidth is the maximum amount of data that can be transferred per second (capacity).

3. Operating Systems (OS)

Q1: What is the difference between a process and a thread? A: A process is an independent program in execution with its own memory space. A thread is a lightweight unit within a process that shares the same memory space with other threads of that process.

Q2: What is deadlock? A: A situation where two or more processes are stuck waiting on each other indefinitely for resources, so none can proceed.

Q3: What is virtual memory? A: A memory management technique that lets a system use disk space to simulate extra RAM, allowing programs larger than physical memory to run.

Q4: What is a race condition? A: When multiple threads/processes access shared data at the same time and the final result depends on unpredictable timing, causing bugs.

Q5: What's the difference between multitasking and multithreading? A: Multitasking = running multiple processes at once (OS-level). Multithreading = running multiple threads within a single process.

Q6: What is a semaphore? A: A synchronization tool used to control access to a shared resource by multiple processes/threads, preventing race conditions.

Q7: What is paging? A: A memory management scheme that divides memory into fixed-size blocks (pages) so a process doesn't need to be stored in contiguous memory, reducing fragmentation.

Q8: What is context switching? A: The process of saving the state of a currently running process/thread and loading the state of another, so the CPU can switch between tasks.

Q9: What is the difference between a mutex and a semaphore? A: A mutex allows only one thread to access a resource at a time (binary lock, ownership-based). A semaphore can allow a fixed number of threads to access a resource simultaneously (counter-based).

Q10: What is thrashing? A: A state where the system spends more time swapping pages in/out of memory than doing actual work, usually caused by too little RAM for the workload.

4. DBMS

Q1: What is normalization? Why is it used? A: Organizing database tables to reduce data redundancy and avoid update/insert/delete anomalies, done through normal forms (1NF, 2NF, 3NF, etc.).

Q2: Explain the different types of SQL JOINS. A: INNER JOIN — only matching rows from both tables. LEFT JOIN — all rows from left table + matches from right. RIGHT JOIN — all rows from right table + matches from left. FULL JOIN — all rows from both, matched where possible.

Q3: What is a primary key vs foreign key? A: Primary key uniquely identifies a row in its own table. Foreign key is a field in one table that references the primary key of another table, creating a relationship.

Q4: What is ACID in databases? A: Atomicity, Consistency, Isolation, Durability — properties that guarantee reliable transaction processing.

Q5: What is indexing, and why does it matter? A: A data structure (often B-tree) that speeds up data retrieval on a table at the cost of extra storage and slower writes.

Q6: What is the difference between DELETE, TRUNCATE, and DROP? A: DELETE removes specific rows (can be rolled back, slower). TRUNCATE removes all rows but keeps the table

structure (faster, usually can't be rolled back). DROP removes the entire table structure and data permanently.

Q7: What is a foreign key constraint used for? A: To maintain referential integrity — ensuring a value in one table actually exists as a valid reference in another table, preventing orphaned records.

Q8: What is the difference between a clustered and non-clustered index? A: A clustered index physically sorts and stores the table's data in that order (only one per table). A non-clustered index creates a separate structure pointing back to the data (a table can have many).

Q9: What is a JOIN vs a subquery — when would you use each? A: A JOIN combines rows from multiple tables in a single query, generally more efficient. A subquery is a query nested inside another, often more readable for filtering based on aggregated results.

Q10: What is database normalization trying to avoid, specifically? A: Redundant data storage and anomalies during insert, update, or delete operations — e.g., updating a customer's address in only one of many duplicate rows, causing inconsistent data.

5. DSA

Q1: What is the time complexity of binary search, and why? A: $O(\log n)$ — because each step eliminates half the remaining search space.

Q2: Difference between an array and a linked list? A: Array: fixed size, contiguous memory, $O(1)$ access by index. Linked list: dynamic size, non-contiguous, $O(n)$ access but $O(1)$ insertion/deletion at a known node.

Q3: What is a hash table, and what's its average lookup time? A: A data structure mapping keys to values using a hash function; average-case lookup, insert, and delete are all $O(1)$.

Q4: Explain the difference between BFS and DFS. A: BFS explores level by level using a queue (good for shortest path in unweighted graphs). DFS explores as deep as possible before backtracking, using a stack/recursion (good for exploring all paths, detecting cycles).

Q5: What is the time complexity of common sorting algorithms? A: Bubble/Insertion/Selection sort: $O(n^2)$. Merge sort/Quick sort (average): $O(n \log n)$. Quick sort worst case: $O(n^2)$.

Q6: What is a stack, and where is it used? A: A LIFO (Last In, First Out) data structure. Used in function call management (call stack), undo operations, and expression evaluation (like checking balanced parentheses).

Q7: What is a queue, and where is it used? A: A FIFO (First In, First Out) data structure. Used in task scheduling, BFS traversal, and handling requests in order (like a print queue).

Q8: What is a binary search tree (BST)? A: A tree where each node's left child is smaller and right child is larger than the node itself, allowing $O(\log n)$ average search, insert, and delete.

Q9: What is dynamic programming, in simple terms? A: An optimization technique that solves complex problems by breaking them into overlapping subproblems, solving each once, and storing (memoizing) the result to avoid recomputation.

Q10: What is the difference between a stack overflow and a memory leak? A: Stack overflow happens when too many function calls (often from deep/infinite recursion) exceed the stack's memory limit. A memory leak happens when a program keeps allocating memory it never releases, gradually consuming available memory.

6. Machine Learning (ML)

Q1: What's the difference between supervised and unsupervised learning? A: Supervised learning uses labeled data (input-output pairs) to learn a mapping, e.g., classification/regression. Unsupervised learning finds patterns in unlabeled data, e.g., clustering.

Q2: What is overfitting, and how do you prevent it? A: When a model learns the training data too well, including noise, and performs poorly on new data. Prevented via more data, regularization, cross-validation, simpler models, or dropout (in neural nets).

Q3: What are precision and recall? A: Precision = of everything predicted positive, how much was actually positive ($TP / (TP + FP)$). Recall = of everything actually positive, how much was correctly predicted ($TP / (TP + FN)$).

Q4: What is the bias-variance tradeoff? A: Bias is error from overly simplistic assumptions (underfitting). Variance is error from being too sensitive to training data (overfitting). Good models balance both.

Q5: What is cross-validation? A: A technique to evaluate a model's performance by splitting data into multiple train/test folds, so results aren't dependent on one particular split.

Q6: What is feature engineering? A: The process of selecting, transforming, or creating input variables (features) from raw data to improve a model's performance.

Q7: What is the difference between classification and regression? A: Classification predicts a discrete category/label (e.g., spam or not spam). Regression predicts a continuous numeric value (e.g., house price).

Q8: What is gradient descent? A: An optimization algorithm that iteratively adjusts model parameters in the direction that reduces the loss/error the most, until it converges to a minimum.

Q9: What is regularization, and why is it used? A: A technique (like L1/L2) that adds a penalty to a model's complexity during training, discouraging overly large weights and helping prevent overfitting.

Q10: What is a confusion matrix? A: A table showing a classification model's actual vs predicted results (True Positives, False Positives, True Negatives, False Negatives), used to calculate metrics like precision, recall, and accuracy.

7. Deep Learning

Q1: What is a neural network, in simple terms? A: A model made of layers of connected nodes (neurons) that learns patterns by adjusting weights based on errors, loosely inspired by the brain.

Q2: What is backpropagation? A: The algorithm used to train neural networks — it calculates the gradient of the loss with respect to each weight by propagating the error backward through the network, then updates weights to reduce error.

Q3: What's the difference between CNN and RNN? A: CNN (Convolutional Neural Network) is designed for spatial data like images, using filters to detect patterns. RNN (Recurrent Neural Network) is designed for sequential data like text or time series, maintaining memory of previous inputs.

Q4: What is an activation function, and why is it needed? A: A function (like ReLU, Sigmoid) applied to a neuron's output to introduce non-linearity — without it, a neural network would just behave like a linear model no matter how many layers it has.

Q5: What is a Transformer, and why does it matter for LLMs? A: A neural network architecture built on “attention” — it lets the model weigh the importance of different words in a sequence regardless of distance. It's the foundation of all modern LLMs like GPT and Gemini.

Q6: What is the vanishing gradient problem? A: When gradients become extremely small as they're propagated backward through many layers, causing earlier layers to learn very slowly or stop learning altogether — common in deep networks with certain activation functions.

Q7: What is dropout, and why is it used? A: A regularization technique that randomly “turns off” a fraction of neurons during training, preventing the network from over-relying on specific neurons and reducing overfitting.

Q8: What is the difference between a loss function and an optimizer? A: The loss function measures how wrong the model's predictions are. The optimizer (like Adam or SGD) is the algorithm that adjusts the model's weights to minimize that loss.

Q9: What is attention, in simple terms? A: A mechanism that lets a model focus more on the most relevant parts of the input when producing an output — e.g., when translating a sentence, attention helps the model know which input word matters most for each output word.

Q10: What is the difference between training and inference? A: Training is when the model learns by adjusting its weights based on data and errors. Inference is when the trained model is used to make predictions on new, unseen input — no weight updates happen during inference.

8. Agentic AI (tied to your projects — expect these to go deeper)

Q1: What is an AI agent, and how is it different from a simple chatbot? A: A chatbot typically just responds to a prompt. An AI agent can reason, make decisions, use tools/APIs, and take multi-step actions autonomously to complete a task — not just generate text.

Q2: What is RAG, and why did you use it in your Document Q&A project? A: Retrieval-Augmented Generation — instead of the LLM answering purely from its training memory, the system first retrieves relevant chunks from the actual uploaded document (via embeddings + similarity search), then feeds that text to the LLM to generate a grounded answer. I used it so answers come from the real document instead of the model guessing.

Q3: Walk me through how your RAG chatbot works end-to-end. A: 1) User uploads a document → 2) it's split into chunks → 3) each chunk is converted into a vector embedding and stored → 4) when the user asks a question, it's also embedded and matched against the stored chunks by similarity → 5) the most relevant chunks are retrieved and passed to the LLM along with the question → 6) the LLM generates an answer grounded in that retrieved text.

Q4: What is LangChain, and what role did it play in your Gemini chatbot? A: LangChain is a framework for building applications powered by LLMs — it handles chaining together prompts, memory, retrieval, and API calls to models like Gemini, so I don't have to manage all that orchestration manually.

Q5: What are vector embeddings, in simple terms? A: A way of converting text into a list of numbers (a vector) that captures its meaning, so that texts with similar meaning end up numerically close to each other — this is what makes semantic search possible.

Q6: How would you handle a case where your chatbot gives a wrong or irrelevant answer? A: Check whether the retrieval step pulled the right chunks (if not, it's a chunking/embedding issue), tune chunk size/overlap, improve the similarity search, or adjust the prompt template so the LLM sticks closer to the retrieved context.

Q7: What is prompt engineering? A: The practice of carefully designing the input/instructions given to an LLM to get more accurate, relevant, or consistent outputs — e.g., telling the model to “answer only using the provided context” in a RAG system.

Q8: Why did you choose Gemini 2.5 Flash specifically for your chatbot? A: It's a fast, lightweight model well-suited for real-time conversational apps, offering a good balance of speed and quality for chatbot-style responses compared to heavier models.

Q9: What challenges did you face while building your resume builder agent? A: Structuring the conversation so it reliably collects all needed details (experience, education, certifications) in the right format, and making sure the generated resume content stayed consistent and well-formatted for different target roles.

Q10: What is the difference between a vector database and a traditional database? A: A traditional database stores and queries exact structured data (rows, columns, exact matches). A vector database stores embeddings and retrieves results based on semantic similarity — finding the “closest meaning” rather than an exact match.

Q11: How would you improve your chatbot projects if you had more time? A: Add better error handling, support more document formats, improve retrieval accuracy with re-ranking, add conversation memory across sessions, and possibly add streaming responses for a smoother user experience.

Quick tip for tomorrow

Interviewers usually spend the most time on whatever's on your resume — so be ready to explain your **RAG chatbot**, **resume builder agent**, and **Gemini chatbot** in your own words, simply, the way it's explained in Q2–Q6 above. That matters more than memorizing every CN/OS definition.